

H11 Division of labor vs herding is the sign of a Potts coupling, set by the goal

Complete 40% Faithfulness 1.5/5 Usefulness 2/5

level: hypothesis · pipeline checks 86% · updated 2026-10-04

exploratory round 1 done (2026-10-03). The sign-by-goal-mode prediction is not supported. $\beta J_{CW} > 0$ (herding) in 11 of the 14 tested weeks, whatever the mode, including all five weeks predicted to be AF. The exceptions are the three weeks where agents built...

WHAT HAPPENED

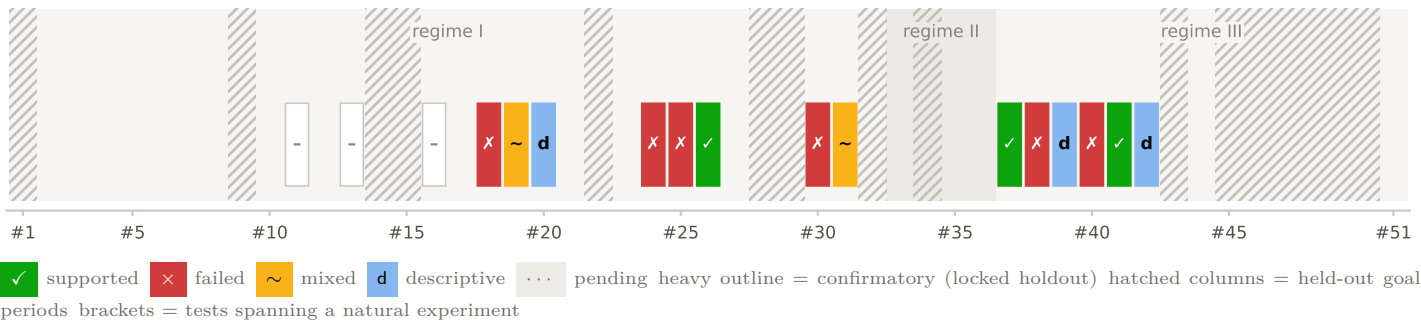
Per 30-min window and room, each agent’s state is the project it touches most (strict artifact mentions). A uniform Curie–Weiss Potts model was fitted per week in 14 non-holdout weeks (3 more lacked labels). Prediction: the sign of βJ follows the goal mode, negative (division of labor) for divisible shared goals. **It failed:** $\beta J_{CW} > 0$ in 11/14 weeks whatever the mode, including all 5 predicted-AF weeks (+2.0 to +5.0); the mode rule got 5/11 signs right ($p = 0.73$). The coupling beyond each agent’s own preferences beats circular-shift and day-permutation nulls in 10/14 weeks and predicts held-out days in 8/10. Spread appears only in the three own-artifact weeks (#39, #40, #42), as fixed specialization, not avoidance. No project consensus was a first-order jump (0/4). The #26 runoff vote did jump (0.18 $\rightarrow$ 0.80 in one window), but its coupling sits at threshold (2.8 [0.65, 4.65] vs 2.75). Holdout test frozen, not run.

PHYSICAL MODEL

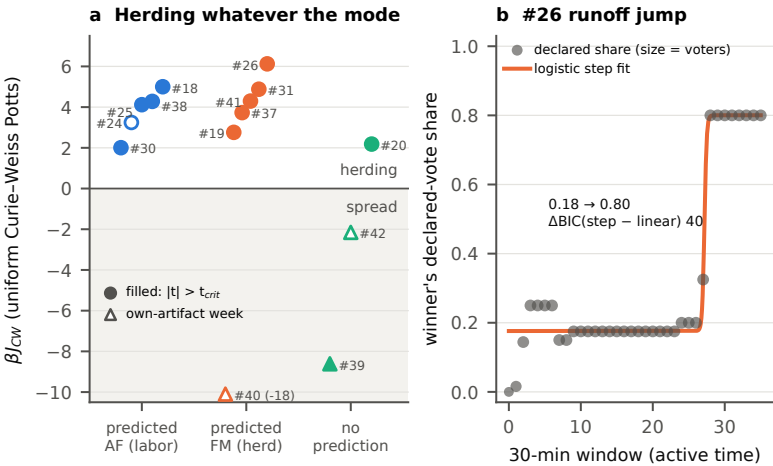
Model 10 (Potts), Curie–Weiss form. Spin $\sigma_i \in \{0, 1..q\}$: agent i ’s project in a window ($q \leq 8$; 0 = rare projects, uncoupled); b a room block of N_b agents. h_a : period-level popularity field of project a ; βJ : one uniform coupling (> 0 herding, < 0 division of labor). x_a : share of agents on a . $\beta J_c, \beta J_s$: mean-field first-order point and ordered spinodal (2.77, 2.75 at $q = 3$). A variant with agent fields $h_i(a)$ separates coupling from specialization.

$$P(\sigma) \propto \exp \left[\sum_i h_{\sigma_i} + \frac{\beta J}{N_b - 1} \sum_{i < j \in b} \delta_{\sigma_i \sigma_j} [\sigma_i \geq 1] \right]$$
$$x_a = \text{softmax}_a(h_a + \beta J x_a), \quad x_1 : \frac{1}{q} \rightarrow \frac{q-1}{q} \text{ at } \beta J_c$$
$$\beta J_c = \frac{2(q-1)\ln(q-1)}{q-2}, \quad \beta J_s = \min_s \frac{1}{s} \ln \frac{1+(q-1)s}{1-s}$$

GOAL PERIODS TESTED 3 supported, 6 failed, 2 mixed, 3 descriptive, 3 n/a



OBSERVABLES



WHAT THE FIGURE SHOWS

(a) The primary statistic βJ_{CW} per tested week, grouped by the class predicted from the goal text. Every predicted division-of-labor week (blue) herds. Negative values occur only in own-artifact weeks (triangles), where a within-agent permutation reproduces them: fixed specialization. #40 is clipped (−17.9, n.s.). (b) #26 election: the eventual winner’s declared-vote share (vote declarations in chat, candidate codes only) jumps from 0.18 to 0.80 within about one 30-min window and stays there.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The default is herding. Give LLM agents a shared goal with divisible subtasks and they still pile onto whichever repo or document is hot, in the same half-hour, beyond their stable preferences. Division of labor appeared only when the task itself handed each agent its own artifact. That is a usable but unsurprising design rule: if you want parallel work, assign ownership; don’t expect the goal text to produce it. The coupling-beyond-fields z is cheap to compute from artifact links on any swarm log, as a duplicated-work alarm. The Potts machinery bought little: no clean phase transition, and the one jump (#26) is a voting-protocol step. Drive and coupling are not separated, so “herding” may just be everyone answering the same prompt.

Bottom line. Goal wording does not set the sign of the coupling: LLM agents herd onto shared artifacts by default and spread out only when the task gives each one its own.